

MAC 233: Servo Lab

Dr Dan Brennan

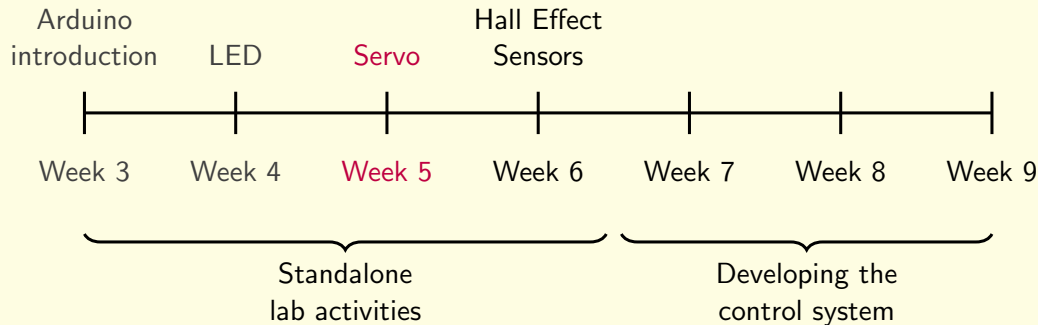
School of Mechanical, Aerospace, and Civil Engineering,
University of Sheffield, UK

MAC 233: **29/10/2025**



University of
Sheffield

MAC 233: LABS OVERVIEW



TODAY'S LAB: PREREQUISITES

This worksheet builds upon the theory of how to use an external pin that was covered in the Lab 2 worksheet. Whilst you won't need any code from Lab 2, you will need to have covered how to use an electronic breadboard and how to wire up an external output to an Arduino pin.

Out of your Arduino pack, you will need: 1 Arduino, 1 Servo, and 3 jumper cables.

TODAY'S LAB: OBJECTIVES

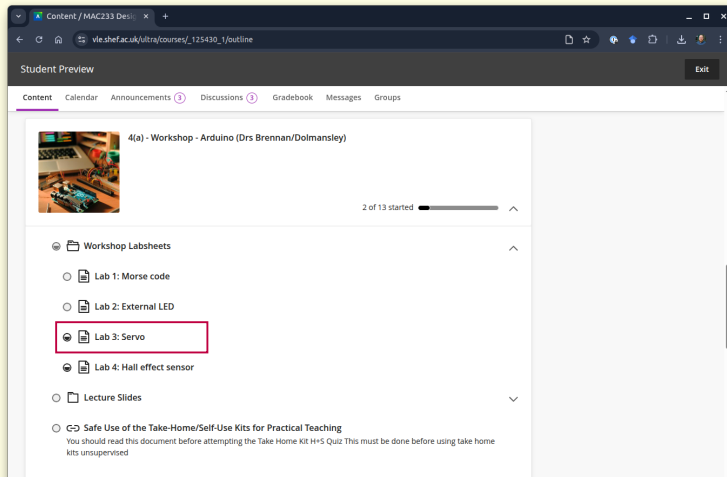
The objective of today's lab is to learn how to control a servo using the inbuilt Arduino Servo Library:

<https://docs.arduino.cc/libraries/servo/>.

The worksheet will take you through the steps of how to connect an external servo to your Arduino and create a sketch that rotates the servo in increments of 30 degrees.

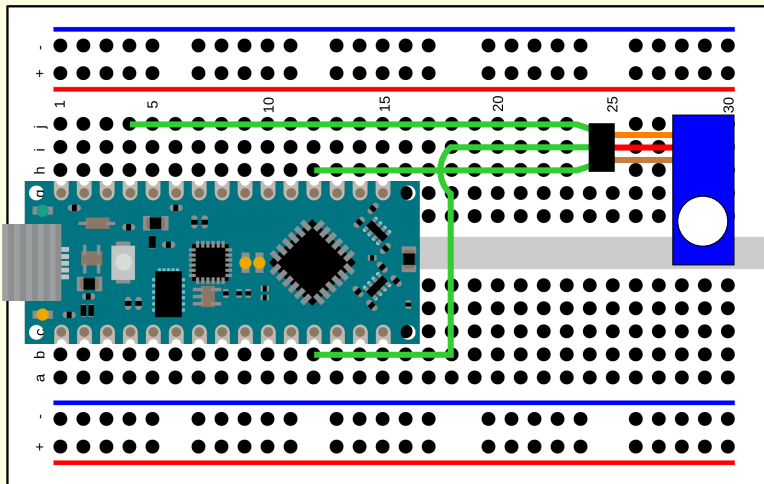
LAB 3: SERVO

Login to Blackboard: <https://vle.shef.ac.uk/>



The screenshot shows a web browser window with the URL vle.shef.ac.uk/ultra/courses/_125430_1/outline. The page is titled "Student Preview" and features a navigation bar with tabs: Content, Calendar, Announcements (3), Discussions (3), Gradebook, Messages, and Groups. The "Content" tab is active, displaying a list of items for the course "4(a) - Workshop - Arduino (Drs Brennan/Dolmansley)". The items include "Workshop Lab sheets", "Lab 1: Morse code", "Lab 2: External LED", "Lab 3: Servo" (highlighted with a red box), "Lab 4: Hall effect sensor", "Lecture Slides", and "Safe Use of the Take-Home/Self-Use Kits for Practical Teaching". A progress bar indicates "2 of 13 started".

WIRING DIAGRAM



Breadboard source: <https://freesvg.org/breadboard>, Resistor source <https://freesvg.org/resistor-330-ohm>.